

v2ecoli → PDMP Whole-Cell Model

Development Roadmap

A six-phase incremental transformation from the current hybrid algorithmic WCM to a piecewise-deterministic Markov process suitable for likelihood-based inference and causal discovery.

Motivation

The current v2ecoli WCM is a hybrid algorithmic simulator combining tFBA optimization, discrete-time stochastic jumps, and teleonomic resource reallocation into a fixed-step loop that does not correspond to any single well-defined mathematical dynamical system. This makes it resistant to likelihood-based inference, causal discovery, and high-performance numerical compilation. The goal of this roadmap is an incremental transformation toward a **piecewise-deterministic Markov process (PDMP)** formulation — combining ODE segments with discrete jump events — without discarding existing biological knowledge.

Each phase replaces one WCM component while keeping everything else intact. At every phase boundary the model must still produce viable *E. coli* growth simulations.

Phase 0 — Characterization and Interface Extraction

- Run v2ecoli in diagnostic mode; categorize every variable as ODE-amenable, stochastic discrete, teleonomic, or coupling artifact.
- Identify the Markov blanket for each major subprocess (metabolism, transcription, translation, replication, division, membrane) and express as typed Process Bigraph interface schemas.
- Deploy the Xarray-format simulation emitter to record full time-series of all interface variables under ≥ 3 environmental conditions.
- Profile v2ecoli for per-step compute time; document marshalling overhead vs. essential computation.

→ **Typed interface schema library + reference Xarray datasets for all process boundaries + profiling report.**

Phase 1 — Metabolism: tFBA → Kinetic ODE

- Implement the Millard et al. (2017) E. coli central metabolism ODE as a self-contained Process Bigraph process using a high-performance ODE library (DifferentialEquations.jl or Diffrax/JAX).
- Wrap with a linear-quadratic regulator (LQR) outer control layer for growth-rate regulation, replacing the tFBA optimization objective with an explicit, parameterized controller.
- Validate using Phase 0 Xarray interface datasets as the context-conditional environment; accept when interface statistics match the reference within tolerance.
- Implement a predictor-corrector shim to integrate the ODE continuously within each v2ecoli timestep, making discontinuities explicit and localized.
- Partition all parameters explicitly as causal (kinetic constants, enzyme abundances) or teleonomic (LQR weights, reference trajectories).

→ **metabolism_ode process + validation report across ≥ 3 conditions + explicit causal/teleonomic parameter partition.**

Risk note: the LQR reference trajectory must not reintroduce a teleonomic parameter in disguise. Parameterization requires careful justification.

Phase 2 — Jump Process Layer: Discrete-Time → Continuous-Time Stochastic

- Replace discrete-time Poisson gene expression draws (currently scaled by timestep/doubling time) with continuous-time jump processes governed by explicit propensity functions from measured kinetic constants (elongation, initiation, degradation rates).
- Replace the teleonomic division trigger (check against target doubling time) with a division event modeled as a first-passage threshold crossing in cell volume or total protein mass.
- Replace the globally acting resource reallocation correction step with competition reactions between processes, implemented as jump events with affinities.
- Couple ODE metabolism (Phase 1) with jump processes as a full PDMP: ODE runs continuously, jump events interrupt and update discrete state (mRNA counts, protein counts, division status).
- Verify the PDMP Markov property: deterministic evolution between jumps, specified probability kernels at jump events.

→ **transcription_jump + translation_jump + division_threshold processes + integrated PDMP runner in Vivarium + trajectory distribution comparison vs. v2ecoli baseline.**

Phase 3 — Inference Infrastructure: Likelihoods and Bayesian Interface

- Specify observation likelihoods for each measurement modality: negative binomial for RNA-seq, log-normal for proteomics, normal on log-growth for growth rate, normal/log-normal for metabolite abundances.
- Implement a likelihood accumulator that runs alongside the PDMP simulator and computes log-likelihood contributions incrementally from ODE segments and jump events.
- Extend Vivarium's algebraic effect system with observe(variable, likelihood) and intervene(variable, value) effects, corresponding to ChiRho's primitives.
- Implement a structured population layer above the single-cell PDMP to compute bulk observables as expectations, bridging to population-level omics likelihoods.
- Deploy ABC-SMC as a baseline inference engine using growth rate, mean mRNA/protein abundances, and metabolite concentrations as summary statistics.

→ **Observation likelihood library + likelihood accumulator + Vivarium observe/intervene effects + ABC posterior recovery on synthetic data.**

Phase 4 — Compilation and Performance

- Export the PDMP WCM to Catalyst.jl / ModelingToolkit for compiled Julia code with symbolic Jacobians, automatic sensitivity equations, and LLVM-optimized simulation.
- Replace nested-dictionary 'row-centric' state with pre-allocated 'column-centric' tensors, enabling vectorized batch simulation with near-zero per-step marshalling overhead.
- Implement the Vivarium API inversion of control: processes declare local state, ODE vector fields, and jump propensities; the runtime handles topology traversal and time-stepping.
- Deploy compiled PDMP runner on GPU / multi-core CPU for $\geq 10^3$ parallel single-cell trajectories; benchmark throughput vs. v2ecoli baseline.

→ **Compiled Catalyst.jl PDMP model + column-centric Vivarium runtime + benchmark report: throughput and ABC wall-clock time.**

Note: Phases 1–2 can proceed in parallel with Phase 4 infrastructure development.

Phase 5 — Causal Discovery and Functional Annotation

- Frame gene function annotation as Bayesian model comparison: compute marginal likelihood of observed data under each candidate annotation using Russian Roulette / doubly-intractable estimators.
- Apply active Bayesian causal inference: propose targeted in-silico interventions (knockouts, overexpression), predict effects under each candidate mechanism, and select experiments that maximally discriminate.
- Validate against a benchmark set of E. coli genes with well-established functions; report posterior recovery rate, credible interval width vs. sample size, and compute cost per annotation.

→ **Bayesian annotation pipeline with credible intervals + active causal discovery demo + full resource estimate for genome-wide scaling.**

Cross-Cutting Principles

- **Incremental replacement.** Each phase replaces one WCM component while keeping the rest of v2ecoli intact. At every phase boundary the model must still produce viable *E. coli* growth simulations.
- **Interface fidelity.** The Markov blanket interface datasets from Phase 0 are the validation standard throughout. A replacement subprocess is accepted when its interface statistics match the reference within a specified tolerance.
- **Causal/teleonomic separation.** Every parameter introduced must be classified as causal (mechanistically derived, inferenceable) or teleonomic (policy/optimality-derived, fixed by design). Inference targets only causal parameters.
- **Forward compatibility.** All process definitions use Process Bigraph schemas and Vivarium API conventions. As the runtime is upgraded in Phase 4, process definitions should require no modification.
- **Documentation as deliverable.** Each phase produces a technical report describing the mathematical model transformation, parameter classification, validation results, and open problems for subsequent phases.

Approximate Resource Requirements

Phase	Primary skills	Effort	Compute
0 — Characterization	Comp. biology, software eng.	2–3 mo (1–2 FTE)	Minimal
1 — ODE Metabolism	Biochemistry, applied math	4–6 mo (2 FTE)	Moderate
2 — Jump Processes	Applied math, stochastic sim.	4–6 mo (2 FTE)	Moderate
3 — Inference	Comp. statistics, comp. biology	6–12 mo (2–3 FTE)	High (ABC/MCMC)
4 — Compilation	Software eng., HPC	4–6 mo (2 FTE)	High (GPU cluster)
5 — Causal Discovery	All above + cell biology	12–24 mo (4+ FTE)	Very high

Phases 0–2 can proceed in parallel with Phase 4 infrastructure development. Phase 3 depends on Phase 2 (the PDMP must exist before likelihoods can be defined). Phase 5 depends on all preceding phases.